DAY 14 - INTERMEDIATE

XLOOKUP & IFERROR

Primary tool: Google Sheets

What you are learning

More flexible lookups and transparent handling of missing keys.

Learning objectives

- Use XLOOKUP

- Handle missing records visibly

- Compare lookup approaches

Worked practical

Replace a VLOOKUP-based solution with XLOOKUP and create an unmatched-record exception count.

Source / Input	Method	Output / Check
Raw data	Apply the lesson skill	Result linked to source
Change one input	Recalculate / rerun	Confirm expected change
Control total / test	Validate	Explain the result

Guided practice

1. Recreate the lesson idea on a small example.
2. Change inputs and predict the result before recalculating.
3. Complete the supplied independent dataset in the required tool.
4. Create a control check or reconciliation.
5. Explain the result in ordinary language.

AI coaching

Ask AI to compare VLOOKUP and XLOOKUP, then verify its claims against Google Sheets behavior.

AI discipline

Attempt first. Use AI for explanation, hints, debugging, critique, test cases and mock questioning. Never submit output you cannot explain and independently verify.

Independent exercise

Create resilient lookup logic.

Common failure modes

- Wrong grain or key.
- Formula/query/code answers a different question.
- Missing or unmatched records are silently ignored.
- Findings are not reconciled to the numbers.
- AI-generated logic is accepted without testing.

Mastery gate

[ ] I can reproduce the method without copying.
[ ] I can trace the result to the source.
[ ] I have a QA/control check.
[ ] I can explain a limitation.
[ ] I can answer an assessor's follow-up question without asking AI.